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SET -5

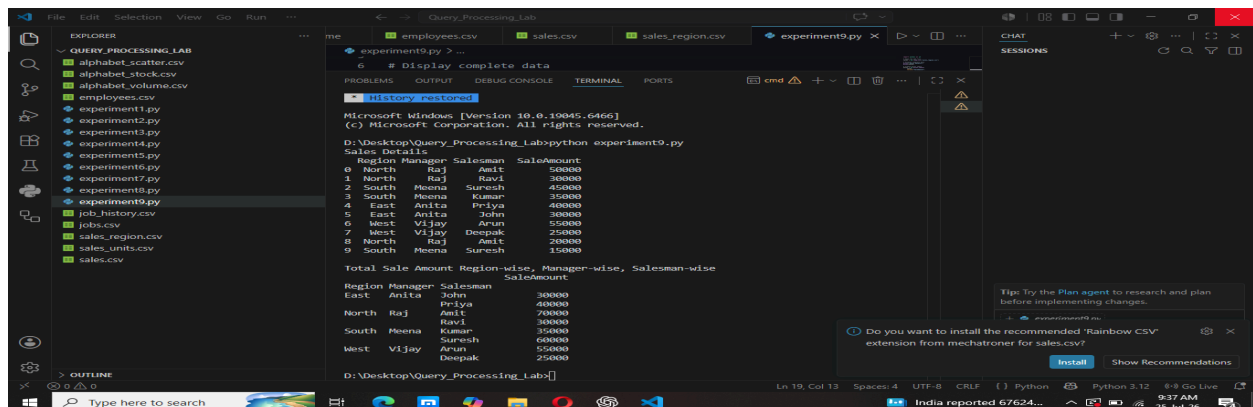
9. Write a Pandas program to create a Pivot table and find the total sale amount region wise, manager wise, sales man wise.

AIM: To Write a Pandas program to create a Pivot table and find the total sale amount region wise, manager wise, sales man wise.

PROCEDURE:

- Import the Pandas library.
- Read the `sales_region.csv` file into a DataFrame.
- Display the sales records.
- Create a Pivot Table using `Region`, `Manager`, and `Salesman` as the index.
- Calculate the total sale amount using the `sum` aggregation function.
- Display the Pivot Table.

OUTPUT:



The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a terminal/output pane at the bottom. The code in the notebook is as follows:

```
6 # Display complete data
```

The output pane displays the following data:

```
Sales Details
Region Manager Salesman SaleAmount
0 North Raj Amit 50000
1 North Raj Ravi 30000
2 South Meena Suresh 45000
3 South Meena Kumar 35000
4 East Anita Priya 40000
5 East Anita John 30000
6 West Vijay Arun 55000
7 West Vijay Deepak 25000
8 North Raj Amit 20000
9 South Meena Suresh 15000

Total Sale Amount Region-wise, Manager-wise, Salesman-wise
Region Manager Salesman SaleAmount
East Anita John 30000
East Anita Priya 40000
North Raj Amit 70000
North Raj Ravi 30000
South Meena Kumar 35000
South Meena Suresh 60000
West Vijay Arun 55000
West Vijay Deepak 25000
```

RESULT: Thus, the Pandas program to create a Pivot Table and find the total sale amount region-wise, manager-wise, and salesman-wise was executed successfully, and the required output was obtained.

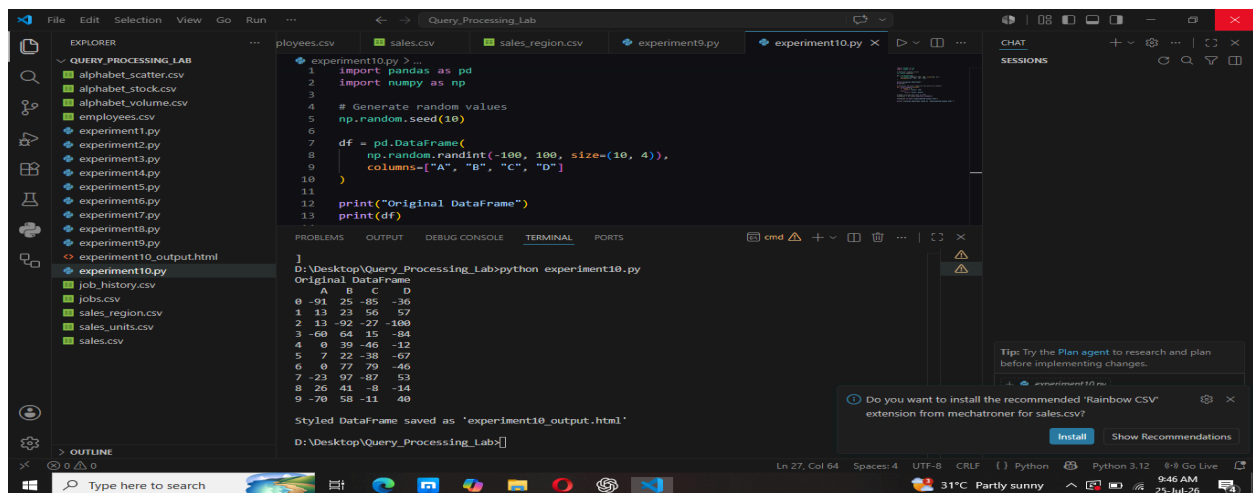
10. Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black.

AIM: To Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black.

PROCEDURE:

- Import the Pandas and NumPy libraries.
- Generate random values with 10 rows and 4 columns.
- Create a DataFrame.
- Display the DataFrame.
- Define a function to color negative numbers red and positive numbers black.
- Apply the style using `style.map()`.
- Save the styled DataFrame as an HTML file.
- Open the HTML file to view the colored output.

OUTPUT:



The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code in the notebook is as follows:

```
1 import pandas as pd
2 import numpy as np
3
4 # Generate random values
5 np.random.seed(10)
6
7 df = pd.DataFrame(
8     np.random.randint(-100, 100, size=(10, 4)),
9     columns=["A", "B", "C", "D"]
10 )
11
12 print("Original DataFrame")
13 print(df)
```

The terminal output shows the original DataFrame:

```
Original DataFrame
   A  B  C  D
0 -91 25 -85 -36
1 13 23 56 57
2 13 -92 -27 -100
3 -60 64 15 -84
4 0 39 -46 -12
5 7 22 -38 -67
6 9 77 79 -46
7 -29 97 -87 53
8 26 41 -8 -14
9 -70 58 -11 40
```

Below the terminal output, a message indicates that the styled DataFrame has been saved as 'experiment10_output.html'.

```
Styled DataFrame saved as 'experiment10_output.html'
```

RESULT: Thus, the Pandas program to create a DataFrame with random values and highlight the negative numbers in red and positive numbers in black was executed successfully, and the required output was obtained.